

ETC3550/ETC5550

Applied forecasting

Week 11: Regression models



Multiple regression and forecasting

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t.$$

- y_t is the variable we want to predict: the “response” variable
- Each $x_{j,t}$ is numerical and is called a “predictor”. They are usually assumed to be known for all past and future times.
- The coefficients $\beta_1, \dots, \beta_k$ measure the effect of each predictor *after taking account of the effect of all other predictors in the model*.
- ε_t is a white noise error term

Uses of dummy variables

Seasonal dummies

- For quarterly data: use 3 dummies
- For monthly data: use 11 dummies
- For daily data: use 6 dummies
- What to do with weekly data?

Uses of dummy variables

Seasonal dummies

- For quarterly data: use 3 dummies
- For monthly data: use 11 dummies
- For daily data: use 6 dummies
- What to do with weekly data?

Outliers

- A dummy variable can remove its effect.

Uses of dummy variables

Seasonal dummies

- For quarterly data: use 3 dummies
- For monthly data: use 11 dummies
- For daily data: use 6 dummies
- What to do with weekly data?

Outliers

- A dummy variable can remove its effect.

Public holidays

- For daily data: if it is a public holiday, dummy=1, otherwise dummy=0.

For monthly data

- Christmas: always in December so part of monthly seasonal effect
- Easter: use a dummy variable $v_t = 1$ if any part of Easter is in that month, $v_t = 0$ otherwise.
- Ramadan and Chinese New Year similar.

Distributed lags

Lagged values of a predictor.

Example: x is advertising which has a delayed effect

x_1 = advertising for previous month;

x_2 = advertising for two months previously;

$\vdots$

x_m = advertising for m months previously.

Fourier series

Periodic seasonality can be handled using pairs of Fourier terms:

$$s_k(t) = \sin\left(\frac{2\pi kt}{m}\right) \quad c_k(t) = \cos\left(\frac{2\pi kt}{m}\right)$$

$$y_t = a + bt + \sum_{k=1}^K [\alpha_k s_k(t) + \beta_k c_k(t)] + \varepsilon_t$$

- Every periodic function can be approximated by sums of sin and cos terms for large enough K .
- Choose K by minimizing AICc or CV.
- Called “harmonic regression”

Comparing regression models

- R^2 does not allow for “degrees of freedom”.
- Adding *any* variable tends to increase the value of R^2 , even if that variable is irrelevant.

Comparing regression models

- R^2 does not allow for “degrees of freedom”.
- Adding *any* variable tends to increase the value of R^2 , even if that variable is irrelevant.

To overcome this problem, we can use *adjusted* R^2 :

$$\bar{R}^2 = 1 - (1 - R^2) \frac{T - 1}{T - k - 1}$$

where k = no. predictors and T = no. observations.

Comparing regression models

- R^2 does not allow for “degrees of freedom”.
- Adding *any* variable tends to increase the value of R^2 , even if that variable is irrelevant.

To overcome this problem, we can use *adjusted* R^2 :

$$\bar{R}^2 = 1 - (1 - R^2) \frac{T - 1}{T - k - 1}$$

where k = no. predictors and T = no. observations.

Maximizing $\bar{R}^2$ is equivalent to minimizing $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{1}{T - k - 1} \sum_{t=1}^T \varepsilon_t^2$$

Akaike's Information Criterion

$$\text{AIC} = -2 \log(L) + 2(k + 2)$$

- L = likelihood
- k = # predictors in model.
- AIC penalizes terms more heavily than $\bar{R}^2$.

Akaike's Information Criterion

$$\text{AIC} = -2 \log(L) + 2(k + 2)$$

- L = likelihood
- k = # predictors in model.
- AIC penalizes terms more heavily than $\bar{R}^2$.

$$\text{AIC}_C = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

- Minimizing the AIC or AICc is asymptotically equivalent to minimizing MSE via **leave-one-out cross-validation** (for any linear regression).

Leave-one-out cross-validation

For regression, leave-one-out cross-validation is faster and more efficient than time-series cross-validation.

- Select one observation for test set, and use *remaining* observations in training set. Compute error on test observation.
- Repeat using each possible observation as the test set.
- Compute accuracy measure over all errors.

Cross-validation

Traditional evaluation

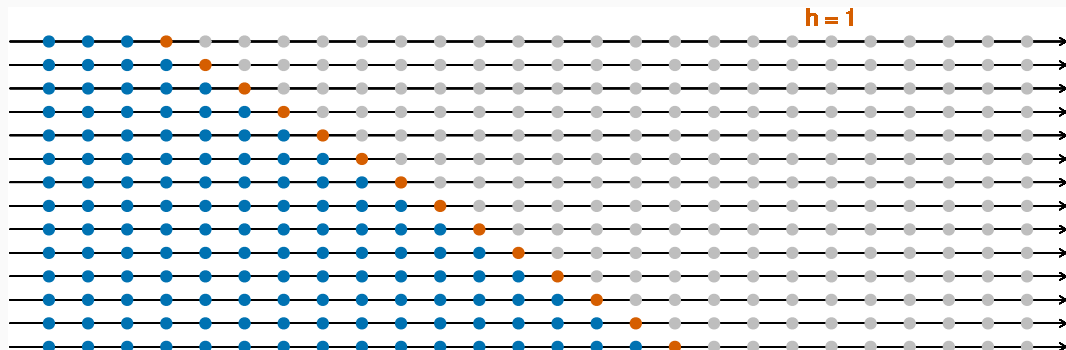


Cross-validation

Traditional evaluation



Time series cross-validation

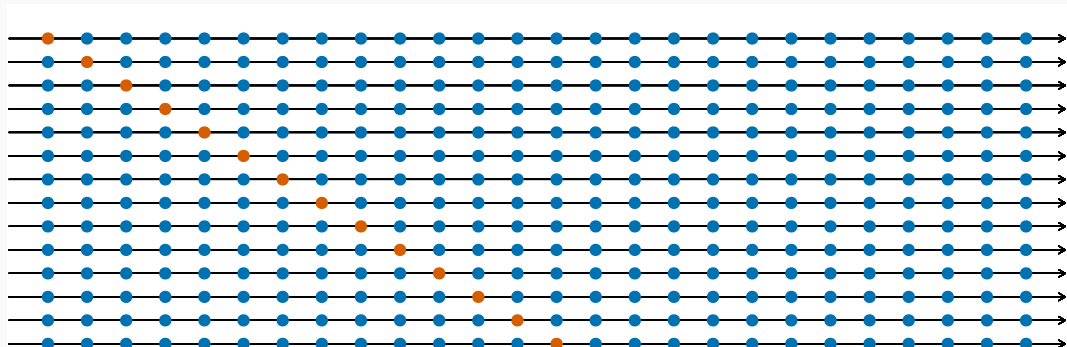


Cross-validation

Traditional evaluation



Leave-one-out cross-validation

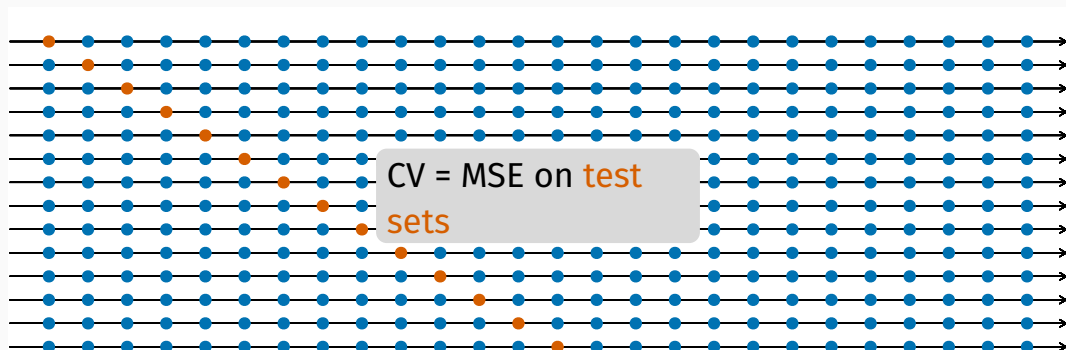


Cross-validation

Traditional evaluation



Leave-one-out cross-validation



Choosing regression variables

Best subsets regression

- Fit all possible regression models using one or more of the predictors.
- Choose the best model based on one of the measures of predictive ability (CV, AIC, AICc).

Choosing regression variables

Best subsets regression

- Fit all possible regression models using one or more of the predictors.
- Choose the best model based on one of the measures of predictive ability (CV, AIC, AICc).

Backwards stepwise regression

- Start with a model containing all variables.
- Subtract one variable at a time. Keep model if lower CV.
- Iterate until no further improvement.
- Not guaranteed to lead to best model.

Ex-ante versus ex-post forecasts

- *Ex ante* forecasts are made using only information available in advance.
 - ▶ require forecasts of predictors
- *Ex post* forecasts are made using later information on the predictors.
 - ▶ useful for studying behaviour of forecasting models.
- trend, seasonal and calendar variables are all known in advance, so these don't need to be forecast.